

Ipsilateral Upper-Limb Lymphedema After COVID-19 Vaccine Booster

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Abstract: Recent data suggest that protective immunity from COVID-19 vaccine may wane after a period, with booster doses becoming increasingly deployed in many countries. An 84-year-old woman presents with a 6-month history of left upper-limb swelling; the only history of note was an intramuscular Pfizer-BioNTech mRNA vaccine booster in the ipsilateral deltoid just 1 week before the onset of the swelling. Upper-extremity lymphoscintigraphy after the administration of ^{99m}Tc -nanocolloid into the interstitial space of the hands showed left upper-limb lymphatic dysfunction.

Key Words: COVID-19, lymphedema, booster, nanocolloid

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FIGURE 1. Grayscale ultrasound image showing thickening of the subcutaneous tissue layer. An 84-year-old woman presents with a 6-month history of left upper-limb swelling; the only history of note was an intramuscular Pfizer-BioNTech mRNA vaccine booster¹ in the ipsilateral deltoid just 1 week before the onset of the swelling. A CT thorax, abdomen, and pelvis done as part of surveillance for her resected gastric cancer were unremarkable.² An ultrasound deep venous thrombosis of the left upper limb shows thickening of the subcutaneous tissue layer but was otherwise unremarkable.

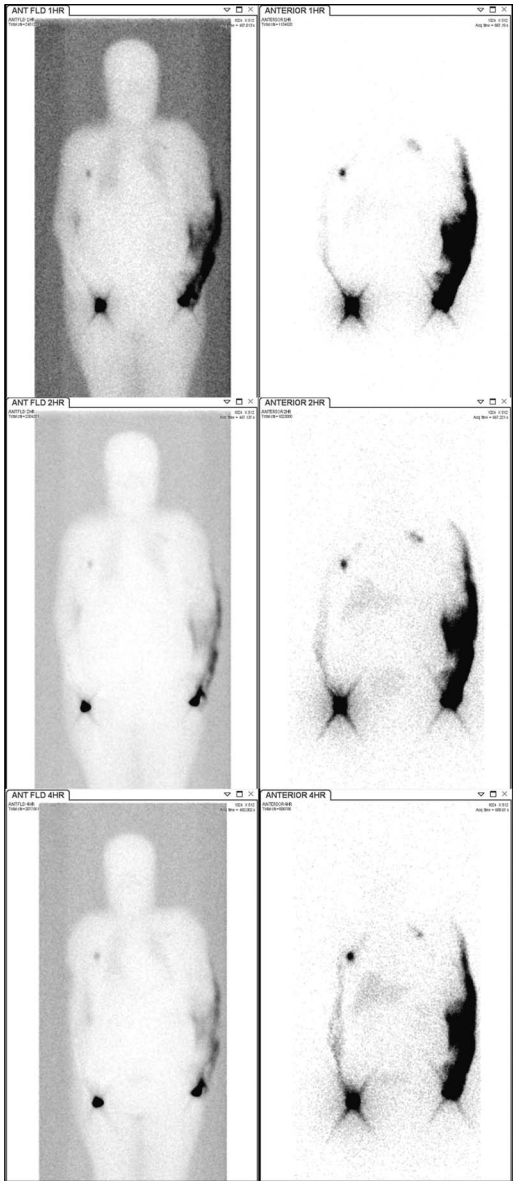


FIGURE 2. An 84-year-old woman presents with a 6-month history of left upper-limb swelling. Upper-extremity lymphoscintigraphy after administration of ^{99m}Tc-nanocolloid^{3,4} into the interstitial space of the hands. Shown are the images 1 hour after injection, images 2 hours after injection, and images 4 hours after injection. Appearance of marked dermal backflow in the left upper limb, with absence of discrete lymphatic channels and absence of radionuclide uptake in expected location of left axillary lymph nodes, suggestive of left upper-extremity lymphatic dysfunction. Lymphadenopathy post-COVID-19 vaccination is now well established and postulated to be related to the transient inflammation from immune system activation.^{5–7} There have also been rare case reports of lower-limb lymphedema after COVID-19 vaccination.⁸ Lymphedema occurs as a result of an imbalance between the lymphatic load and the transport capacity of the lymphatic system.⁸ In this patient, the vaccination may have resulted in damage to the lymphatic vessels or lymph nodes with a decrease in the numbers or obstruction of lymph channels.